

Product Requirements Document (PRD)

Nurture Roots Hour of AI Platform

Version: 1.0

Prepared For: Nurture Roots Foundation

Programme: Nurture Roots Code Club – Hour of AI

Document Type: Product Requirements Document (PRD)

1. Executive Summary

The **Nurture Roots Hour of AI Platform** is a web-based volunteer orchestration and learning coordination platform designed to support a global AI learning programme for children and young people.

The platform connects:

- Volunteer mentors (primarily Microsoft employees and global professionals)
- Remote learners worldwide
- Learners attending physical community learning hubs in underserved regions

The system automates learner and mentor registration, class allocation, volunteer scheduling, session coordination, impact tracking, and volunteer hours reporting.

The platform will support both online learners and physical hubs while ensuring structured mentor coverage, reliable session delivery, and scalable programme operations.

2. Background

Nurture Roots Foundation currently runs the **Hour of AI**, a weekly global learning initiative focused on introducing children and young people to Artificial Intelligence, coding, and digital literacy.

Sessions are delivered virtually through:

- Microsoft Teams
- Code.org learning resources

The programme operates using volunteer mentors and supports learners from diverse global locations, including underserved rural communities through physical learning hubs.

Current coordination activities are largely manual, creating challenges in:

- Scheduling
- Availability matching
- Session management
- Volunteer coordination
- Impact reporting
- Mentor replacement handling

The platform aims to solve these operational challenges and enable global scaling.

3. Product Vision

To build a scalable global platform that enables:

- Efficient coordination of volunteer-led AI education
 - Automated learner and mentor matching
 - Reliable delivery of weekly sessions
 - Real-time impact measurement
 - Seamless support for both remote learners and physical hubs
-

4. Product Objectives

The platform objectives are to:

1. Automate mentor and learner registration
 2. Match mentors and learners automatically
 3. Reduce manual operational overhead
 4. Ensure reliable mentor coverage for every class
 5. Track volunteer hours and programme impact
 6. Support underserved communities through physical hubs
 7. Improve learner retention and engagement
 8. Enable future scalability across multiple programmes
-

5. User Types

5.1 Students / Learners

Children and young people aged 8–18 participating in the programme.

Types

- Remote learners
 - Hub learners
-

5.2 Volunteer Mentors

Professionals supporting sessions voluntarily.

Primary Mentor Sources

- Microsoft volunteers
 - Global professionals
 - Community mentors
-

5.3 Administrators

Programme management personnel responsible for:

- Monitoring operations
 - Managing registrations
 - Reviewing assignments
 - Handling safeguarding issues
 - Monitoring impact metrics
-

5.4 Hub Coordinators

Local representatives responsible for managing physical learning hubs.

Responsibilities include:

- Supervising learners
 - Supporting attendance
 - Coordinating local operations
-

6. Scope

In Scope

- Student registration
 - Volunteer registration
 - Hub registration
 - Availability management
 - Automated class assignment
 - Session management
 - Microsoft Teams integration
 - Volunteer hours tracking
 - Mentor absence handling
 - Replacement coordination
 - Notifications and reminders
 - Impact reporting
 - Admin dashboard
-

Out of Scope (Phase 1)

- AI-powered scheduling optimization
 - Gamification
 - Mobile applications
 - Advanced analytics
 - Video recording storage
 - Automated Code.org progress synchronization
-

7. Functional Requirements

7.1 Landing Page

The platform shall provide a public landing page containing:

- Programme overview
- Registration options
- Impact metrics
- Information for:
 - Students
 - Volunteers
 - Community hubs

Primary Actions

- Register as Student
 - Register as Volunteer
 - Register a Hub
-

7.2 Registration Module

7.2.1 Student Registration

The system shall allow students to register using an online form.

Required Information

- Full name
- Age
- Country
- City
- Timezone
- Participation type
- Parent/guardian email
- Weekly availability

Participation Types

- Remote learner
- Hub learner

If hub learner:

- Select associated hub
-

7.2.2 Volunteer Registration

The system shall allow mentors to register.

Required Information

- Name
- Email
- Organization
- Country
- Timezone
- Preferred learner group
- Weekly availability
- Maximum sessions per week

Optional Information

- Skills
 - Experience level
-

7.2.3 Hub Registration

The system shall allow physical learning hubs to register.

Required Information

- Hub name

- Location
 - Coordinator details
 - Device availability
 - Internet connectivity quality
-

7.3 User Categories

The system shall automatically assign learners into categories based on age.

Category	Age Range
Juniors	8–10
Intermediates	11–14
Seniors	15–18

Classes shall only contain learners from the same category.

7.4 Availability Management

The platform shall allow:

- Students to manage availability
- Mentors to manage availability
- Recurring weekly schedules
- Temporary unavailability updates

Availability shall include:

- Day of week
 - Start time
 - End time
 - Timezone
-

7.5 Assignment Engine

The platform shall automatically generate class assignments.

Assignment Rules

Mentor Rules

Each class must include:

- 1 primary mentor
- 1 backup mentor

Minimum:

- 2 mentors per class
-

Student Rules

Each class must contain:

- Minimum 3 students
- Maximum 5 students

Matching Criteria

Assignments shall be based on:

- Availability overlap
- Learner category
- Mentor workload capacity
- Timezone compatibility

Class Types

Classes may be:

- Remote
- Hub-based
- Mixed

Preference shall be given to grouping hub learners together.

7.6 Session Management

The platform shall generate recurring weekly sessions for approved classes.

Each session shall include:

- Schedule
- Mentors
- Students
- Microsoft Teams link
- Code.org lesson reference
- Attendance
- Session notes

7.7 Microsoft Teams Integration

The platform shall integrate with Microsoft Teams to:

- Generate recurring meetings
- Share join links
- Notify mentors and learners

7.8 Volunteer Hours Tracking

The platform shall track volunteer hours delivered.

Features

- Session duration tracking
- Manual hours logging
- Reminder notifications
- Total hours reporting

Mentors shall receive reminders after sessions to log volunteer hours.

7.9 Mentor Absence Management

Mentors shall be able to submit absence requests.

Types

- Planned absence
 - Emergency absence
-

Replacement Workflow

When a mentor is unavailable:

1. Backup mentor becomes primary mentor
 2. Replacement request is broadcast to all mentors
 3. First eligible mentor to accept is assigned automatically
-

Emergency Absence Handling

Emergency absence notifications shall be sent to:

- Administrators
 - Hub coordinators
 - All mentors
-

7.10 Notifications

The system shall support automated notifications for:

Session Reminders

- 24 hours before session
- 2 hours before session

Operational Notifications

- Assignment confirmations
- Replacement requests
- Hours logging reminders

Notifications shall initially be delivered via email.

7.11 Impact Tracking

The platform shall provide impact metrics including:

- Total learners
 - Total mentors
 - Volunteer hours delivered
 - Sessions delivered
 - Countries represented
 - Active hubs
-

7.12 Admin Dashboard

Administrators shall be able to:

- Approve registrations

- Review assignment proposals
 - Manage sessions
 - Monitor volunteer hours
 - Review impact metrics
 - Handle mentor replacements
 - Manage safeguarding concerns
-

8. Non-Functional Requirements

Scalability

The platform shall support:

- Multiple countries
 - Multiple hubs
 - Hundreds of concurrent users
-

Security

The platform shall implement:

- Secure authentication
 - Role-based access control
 - Secure data storage
-

Safeguarding

The platform shall include:

- Parent consent collection
 - Incident reporting
 - Role restrictions
-

Reliability

The system shall:

- Prevent scheduling conflicts
 - Handle mentor replacement efficiently
 - Ensure data consistency
-

9. Technical Requirements

Frontend

- React / Next.js
-

Backend

- Node.js / Serverless Functions
-

Database

- PostgreSQL or CosmosDB

Authentication

- Microsoft Entra ID
- Email authentication

Integrations

- Microsoft Teams
- Microsoft Graph API
- Code.org
- Email service provider

10. Success Metrics

The platform will be considered successful if it achieves:

- Reduced manual scheduling effort
- Increased session delivery reliability
- Increased volunteer retention
- Increased learner participation
- Accurate volunteer hour tracking
- Expansion to additional global hubs

11. Future Enhancements (Post-MVP)

Potential future enhancements include:

- AI-powered scheduling optimization
- Mentor reliability scoring
- Attendance analytics
- Gamification and badges
- Mobile applications
- Automated Code.org progress synchronization
- Volunteer leaderboards
- AI-generated impact reports

12. Risks and Dependencies

Risks

- Timezone coordination complexity
- Volunteer availability fluctuations
- Connectivity limitations in rural hubs

Dependencies

- Microsoft Teams availability
- Microsoft Graph API
- Reliable email delivery service

- Internet connectivity at hubs
-

13. Conclusion

The Nurture Roots Hour of AI Platform will provide the operational infrastructure required to scale a global volunteer-led AI learning initiative efficiently and sustainably.

The platform will enable:

- Automated coordination
- Reliable session delivery
- Scalable volunteer management
- Structured impact reporting
- Enhanced access to AI education globally

while supporting both remote learners and underserved communities through physical learning hubs.